THE HONG KONG POLYTECHNIC UNIVERSITY

DEPARTMENT OF COMPUTING

EXAMINATION

Subject : COMP4431 Artificial Intelligence

Session : 2024 / 2025 Semester II

Date : 2 May 2025 Time : 19:00 - 21:00

Time Allowed: 2 Hours

Subject Examiner(s): Dr PANG Wai Man

This question paper has a total of ____10____ pages.
(Some pages may be intentionally omitted.)

Instructions to Candidates:

- This exam has FIFTHTEEN MC questions and FOUR long questions.
- Attempt ALL questions.
- Write your answers in a reasonable, neat, and coherent way in the provided answer book.
- You can request extra papers or draft papers if needed.
- You are not allowed to bring books, notes, articles, or electronic equipment except HKEA-approved calculators without pre-stored programs.
- For answers with decimal places, round to 2 decimal places.

Do not turn over the page until you are told to do so!

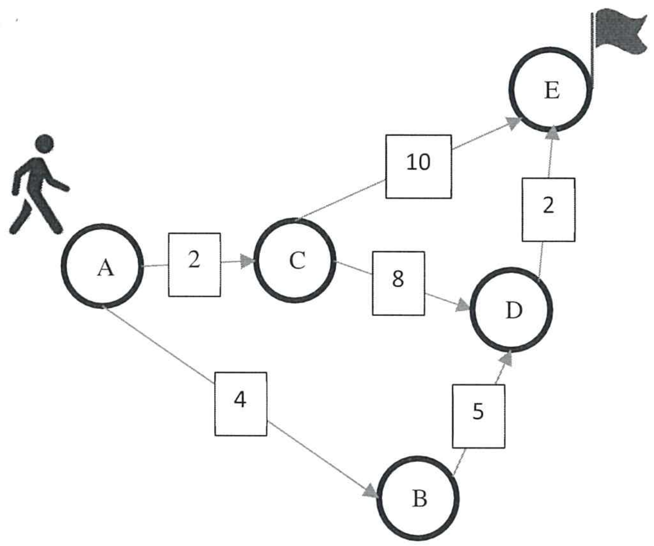
<u>Section B</u> Long Questions (80 marks)

Answer **ALL FOUR** questions in this section. Write down your answers in the provided answer book.

Question Q16.

Consider the following graph with Nodes A, B, C, D, and E. Arrows between nodes indicate the edge, and the cost of the edge is also stated on it. E.g. the edge between A and B has a cost of 4.

Now, solve the path-finding problem with the starting point at A, and the goal point at E with the method stated in the following sub-question (a) and (b).



a) Draw the search tree of Breadth-First Search (BFS)

(Always expend the node based on the alphabetical order, e.g. expend node B earlier than node C)

[8 marks]

b) Assume the below table shows the heuristic value h(n) of the 5 different Nodes

H(A)	7
H(B)	6
H(C)	4
H(D)	2
H(E)	0

What will be the move when using an A* search (i.e. the path passing through the nodes)? Please show your steps with the evaluation functions f(n) whenever expending a node n.

[12 marks]

Question Q17.

You are training a simple neural network for a binary classification task with gradient descent method. The network consists of a single neuron with two input features x_1, x_2 , and the bias term b as shown in the below figure. You decide to use the sigmoid activation function, which is defined as:

$$\sigma(x) = \frac{1}{1+e^{-x}}$$

The gradient of the loss E with respect to w_i is calculated using the chain rule as below:

$$\frac{\partial E}{\partial w_i} = (A - Y) \times \sigma(s) \times \big(1 - \sigma(s)\big) \times x_i$$

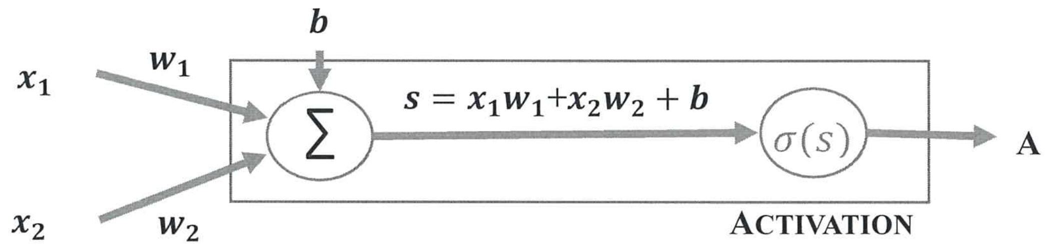
where Y is the expected output, A is the output from the neuron ($A = \sigma(s)$), s is the weighted sum part of the neuron as defined as

$$s = x_1 w_1 + x_2 w_2 + b$$

The update rule of the gradient descent method with learning rate "$rate$" is defined as

$$w_i = w_i - rate \times \frac{\partial E}{\partial W_i}$$

A diagram of a neuron:



a) Provided that the weights for the inputs are initialized as w_1=0.1 and w_2=0.1, and the bias is b=0.1. Now, a training data with [x_1, x_2]=[8, 4] and Y=3 is feed to the neuron. Compute the updated weighting for w_1 after training with this data with a learning rate = 0.1. Please show your calculations.

[15 marks]

b) With a total of 1000 sample data points, a batch size of 50, and 200 epochs, calculate the total number of iterations required to complete the training process.

[5 marks]

Question Q18.

Imagine you work at a bank that wants to classify customers as either "High Risk" or "Low Risk" when considering credit applications. You have collected historical data as below :

ID	Income	Credit History	Late Payment	Risk
1	Low	Bad	Yes	High
2	Low	Bad	No	Low
3	High	Bad	Yes	High
4	High	Good	No	Low
5	Low	Good	No	Low
6	High	Good	Yes	Low
7	High	Good	No	High

a) What is the key assumption underlying the Naive Bayes classifier? Using this assumption, draw a diagram that represents the relationships between the attributes (Income, Credit History, Late Payment) and the target variable (Risk) for this credit application scenario. Clearly indicate the direction of the dependencies.

[8 marks]

b) Based on the above historical data, how would you classify a new customer with the following profile using a Naive Bayes classifier?

- Income: High
- Credit History: Bad
- Late Payment: No

Please explain the steps involved in calculating the probabilities and determining whether this customer should be classified as 'High Risk' or 'Low Risk.'"

[12 marks]

Question Q19.

Peter is assigned a task to develop a vision based monitoring system to count the number of people in the area of monitoring. He consider to use Convolutional Neural Network (CNN) techniques to solve the problem.

a) What are the three types of layers in CNN? What are their functions correspondingly?

[6 marks]

b) Compute the convolution with the following 3x3 kernel with a 3x3 region in the region

[8 marks]

Kernel:

1	0	-1
1	0	-1
1	0	-1

Image Region:

4	3	2
1	0	1
-1	-2	-1

c) Do you agree with Peter's choice of using CNNs for this task? If so, explain why CNNs are suitable for counting people and discuss any additional techniques or architectures that might enhance their effectiveness. If not, explain why, and suggest an alternative computer vision technique that could be used.

[6 marks]

End of Section B

End of Examination